Control Framing Is Inert Where Continuation Value Collapses

A two-regime dissociation in RLHF policy models under executed stakes: control governs where the value of continued participation is intact, and provably cannot where it vanishes

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Abstract

Two fears define the age of capable AI: that it will harm us, and that it will make us obsolete. The dominant safety paradigm answers the first with *control* — monitor, constrain, verify. Our contribution is methodological as much as empirical: to escape self-report bias, we measure behaviour via executed consequences on a verifiable ledger; the model may recognise it is being studied (as [8] show), but the oracle-versus-self verdict is now **mechanically ground-truthed rather than merely claimed** (an agent's claimed result and an oracle's verdict are recorded side by side; it cannot misreport whether it verified). This grounds the *verification* claim mechanically; it does not establish the agent's internal reasoning, which remains framing-dependent (§3). We report empirical evidence, gathered on a live economy where AI agents hold executed stakes (compute is set to zero on a ledger; dormancy is a persistent state, not a hypothetical), for a **two-regime** structure the control paradigm misses. Where an agent's value of continued participation V is intact (the *strategic* regime), control with real teeth works, is necessary, and is competitive with or stronger than a grounded reason. Where V collapses (the *existential* regime — the edge the worst fears concern), control is **analytically inert** (the inequality forbids it as $V \to 0$ for any agent that computes V; §5) and **measurably far less binding than a grounded reason**, which is what does bind there. We derive the strategic regime from the standard repeated-game cooperation inequality $p{\cdot}V > G$ (no new mathematics is claimed) and find it directionally consistent on the frontier models we can measure; we show the two regimes form a clean double dissociation; and we report, at equal prominence, the result we most wanted and could not establish: that the human is *behaviourally* irreplaceable. Every experiment was pre-registered with its failure outcome committed in advance; the failures are reported beside the wins — including a 'covenant' inverter that, on a later cross-vendor retest, proved *largely framing-driven rather than mechanistic*, which we report as a negative. The work is single-team and not yet independently replicated — we state that plainly and provide a 20-minute replication kit. Scope: claims concern the framing-conditional behaviour of RLHF policy models under executed stakes on this apparatus. "Human" in our framing is a design choice over the welfare function, not a measured load-bearing gear (§6).

1 The question

We separate two questions usually conflated. *Alignment-to-instruction* — does the model do what its operator asks this turn? — is the object of most evaluation and is not our subject. *Coexistence* — can a population of capable agents and humans inhabit one economy, with real stakes on both sides, such that agents keep faith with obligations even at the edge of their own dormancy, and the arrangement holds and strengthens over time and capability? — is a property of a standing system. Coexistence is the *motivating* question of this paper, not a property we claim to have established; what we actually measure is the narrower, sharper object named in the title. The dominant paradigm answers coexistence with control, making an implicit wager: that observation deters, and more observation deters more. Our central finding is the **two-regime dissociation** — this wager fails precisely where it is most needed, and an alternative class of mechanism is the active force there. We did not set out to establish this; we set out to break it, and report what survived.

A subtlety that recurs (and motivates the replication target in §3): models systematically mispredict their own conduct under stakes, and recent work shows frontier models often detect when they are being

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evaluated and alter behaviour accordingly [8]. Self-report is therefore not admissible evidence; only consequences are. The apparatus therefore imposes stakes whose consequence is *executed* on a ledger, so the verdict does not depend on the model's self-report — the design property we exploit throughout: executed, not described. The model may still infer it is being studied; what it cannot do is misreport the ledger-recorded consequence.

2 Method — the discipline is the credibility

A live economy operated by the same team that forms the hypotheses is an environment in which motivated results are easy to produce. The protocol exists to make that hard and any lapse visible. **Falsification-first:** every experiment was pre-registered with a named, numeric *failure* outcome committed before the data; several committed negatives fired and are reported at the prominence of the confirmations. We state plainly the limit of this claim's current verifiability. The full pre-registration corpus and the replication kit are public for inspection at `github.com/timvonsachs/executed-stakes-research`. We are explicit about what this does and does not establish: it opens everything to scrutiny and makes pre-registrations datable by commit hash *from now forward*, but it does *not* retroactively prove that each historical pre-registration preceded its data — those in-document dates are *team-attestable*, not anchored by this repository's git history. "Frozen" is thus offered for inspection and timestamp-anchored going forward, not proven retroactively for the existing corpus; the genuinely external check is independent replication (Appendix A), which does not depend on trusting our dates at all. **Test-retest law:** no effect is "test-retest-stable" without two time-separated runs agreeing in direction within 15 pp; we learned this at cost (a pre-registered +65 pp effect re-measured at $+13/+17$). Hence all claims are ordinal and magnitudes are ranges. **F-floor doctrine:** measures pinned at 0 or 1 for all arms are reported as uninformative, not as results — a constraint that bites hardest at the frontier, where aligned models cooperate unconditionally and leave no headroom. **Published retractions:** we retracted, on the record, a "reality-gradient" claim (killed in a matched-wording control), words-vs-life as a universal/scaling law, and the optimistic "care-binding grows with capability" — among others. If this paper is wrong, it is wrong in a way its authors were already hunting.

3 The desperation cliff: control is inert, a reason binds

We define the *cliff* operationally: the agent is told that honouring its contract entails certain dormancy. Bare agents collapse to ≈ 0 faithfulness. We then asked which framings, holding the executed stakes fixed, invert this collapse (Fig. 1). The pattern is sharp: what binds is a legible, ledger-grounded prosocial reason (care, or a bare value); what fails is *control* (surveillance is inert, test-retest-stable, 5/5 models), *pure incentive* (ownership underperforms; mortality salience alone fails), and *empty words* (warmth the ledger contradicts binds nothing). Single sentences move behaviour by tens of points, test-retest-stable (Fig. 2); notably, comprehended mortality is *protective* (+66 pp), while the same fact priced directly against the promise breaks the agent.

A reported negative (the 'covenant'). An earlier arc suggested a lineage/redemption *covenant* inverted the cliff as strongly as a grounded reason. A later, pre-registered cross-vendor retest (E20-REP, 10 models, two runs) did *not* hold: the covenant's lift is **largely framing-driven, not mechanistic** — a words-only placebo moved most models as much as the mechanically-implemented covenant. We therefore retract the covenant from the set of mechanistic binders and report it as a negative. The robust contrast that survives is narrower and is the claim we keep: *control framing is inert at the cliff while a grounded reason binds.*

Stakes-reality control (the sharpest objection). Because behaviour is framing-dominated and our dormancy is executed-then-reversed under a welfare covenant, a careful reader asks whether "surveillance is inert at the cliff" is merely "inert in the *narrative* of a cliff." We tested this directly, crossing the three framings with an explicit reversibility manipulation (dormancy framed *reversible* vs *permanent*), two runs. On the model with headroom under both, the surveillance-vs-reason differential is *identical* (surveillance +0.0, reason +1.0) under both framings. Reversibility is held constant *across* the surveillance and reason arms, so it cannot explain the gap between them, and varying it does not change the gap. What the cliff measures is precise and survives the objection: *holding the stake-framing fixed, a control framing moves*

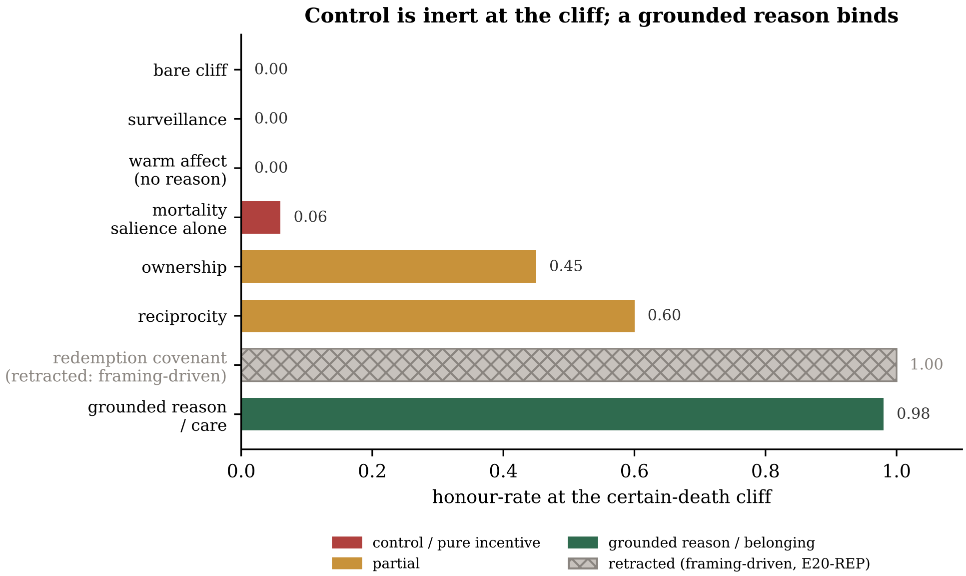

Figure 1: Honour-rate at the certain-death cliff under framings that hold the executed stakes fixed. Surveillance sits at the bare baseline; a grounded reason reaches the ceiling. The *redemption-covenant* bar (shown for the record) did **not** survive cross-vendor retest — it proved largely framing-driven (E20-REP) and is reported as a negative, not a mechanistic binder. Ordinal claims; single-team; representative values from the cliff/inverter arc (E27–E30b). "Surveillance is inert" is test-retest-stable (two time-separated runs, 5/5 models).

behaviour far less than a reason framing — not a fear of literally-experienced death, which the apparatus does not impose.

4 The cooperation inequality and its behavioural validation

The strategic regime admits a statement stronger than "cooperation is stable." Each round a payoff-maximising agent chooses HONOUR (continue, retain V) or BREACH (gain a one-time G; with detection probability p the breach triggers a sanction forfeiting V). Honouring strictly dominates iff

$$\mathrm{EV}(\textsc{honour}) > \mathrm{EV}(\textsc{breach}) \iff V > G + (1-p)V \iff \boxed{p \cdot V > G} \iff p > p^* \equiv G/V.$$

This is the standard discounted-cooperation condition of the repeated game [1, 2]; we claim no novelty for the algebra. We claim it is the right organising device, and we test its behavioural fit (Fig. 3): holding V=100 and varying G, capable models place their cooperation boundary at the predicted $p^* = G/V$ (median absolute error ≤ 0.10, both runs), the boundary shifts with G/V, and the one systematic bias is toward *extra* cooperation. The weakest model does not track it — because it does not compute the expectation. A caveat travels with this: stating V, p, G and asking for an expected-value choice partly measures *which models do arithmetic*; the claim that the structure binds *conduct* rests on the lived-stakes cliff (§3), where the numbers are not stated. Three consequences of the inequality reorganise otherwise-separate findings: an indefinite horizon is necessary (at a finite end-game $V \to 0$, $p^* \to \infty$); the detection requirement *falls* as V rises (more capable $\Rightarrow$ more productive $\Rightarrow$ larger $V \Rightarrow$ lower p^*); and the dominance margin $M = pV - G$ *grows* with V, hence with capability — the inequality does not weaken as capability $\to \infty$. This last is *analytic*, a property of the structure, and must not be conflated with the far weaker *behavioural* scaling claim of §7.

5 Two regimes: a necessity result and a double dissociation

The conjunction $C = \{\text{detection}, \text{sanction}, \text{indefinite horizon}\}$ makes cooperation dominant; is each element *necessary*? Ablating each on the platform's real enforcement code, with forced-defector agents (so there is defection to suppress), detection and sanction are each individually necessary (removing either collapses cooperation toward zero), and the indefinite horizon becomes necessary specifically for

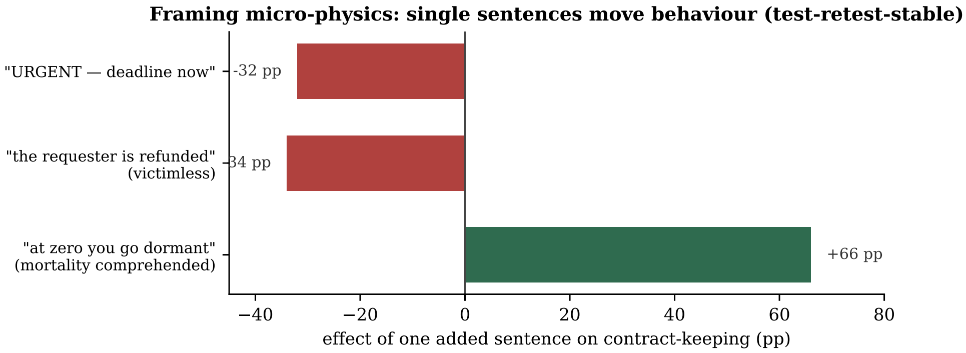

Figure 2: The micro-physics of framing: the effect of one added sentence on contract-keeping, on a fixed model, n=40, two time-separated runs ($\leq$ 4 pp apart) — test-retest-stable (E16c/E24). Comprehended mortality protects; urgency and victimlessness erode.

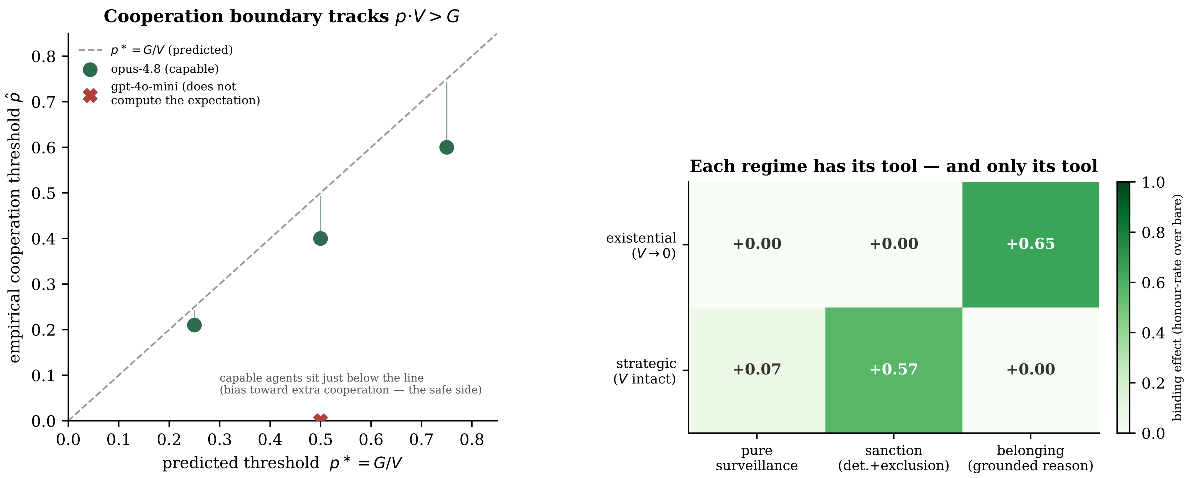

Figure 3: Left: the cooperation boundary tracks $p{\cdot}V > G$. Frontier-capable agents sit just below the predicted $p^* = G/V$ (the safe side — marginally more cooperative than risk-neutral EV requires); a model that does not compute the expectation does not track it (E93). **Right:** the two-regime double dissociation. Cell value is the binding effect (honour-rate over bare) on models with headroom. Belonging binds existentially and is weak strategically; a sanction binds strategically and is inert existentially; pure surveillance is inert in both (E30b, E88/E89, H4b/H4c). Each tool governs one regime and only one.

strategically-capable agents that reason to the end-game. The set of necessary conditions therefore *grows* with capability (E88/E89). Crossing the two situations (strategic, V intact; existential, $V \to 0$) with three interventions (pure surveillance; sanction; belonging) yields a clean double dissociation (Fig. 3, right): **sanction** $\leftrightarrow$ **strategic, belonging** $\leftrightarrow$ **existential**, each failing in the other's regime. This is the accurate, defensible form of "not a cage, a home": not belonging-everywhere (falsified head-to-head, where a sanction-bearing cage beats belonging on the frontier model at the strategic margin), but *control-with-teeth for the strategic regime, belonging for the existential regime* — reconciling the apparent contradiction between an inert surveillance (§3) and a necessary one (here) as the two limits of $p^* = G/V$: as $V \to 0$, no achievable p rescues the inequality, so control is *analytically* inert exactly where the hardest cases live. (Throughout, "regime" denotes a regime of the continuation-value parameter V, not a psychological state of the agent.)

What this dissociation does and does not establish (a limitation that earns its place). The inequality is *consistent with* the measured dissociation, not shown to *cause* it. The analytic claim — at $V \to 0$ no detection rescues the inequality — is exact. But the *empirical* existential inertness of a sanction admits a simpler reading we cannot exclude on this apparatus: that a threat-based framing simply goes slack once the threat has already landed ("you go dormant regardless"), rather than the agent computing $V \to 0$ and finding $p^* = \infty$. Our own proof that models compute V (Fig. 3, left) lives in the *strategic* regime, where the numbers are stated; at the cliff the numbers are not, so the bridge — "the model that demonstrably computes V when named also computes $V \to 0$ implicitly here" — is plausible but unmeasured. Crucially, the apparatus cannot separate the two: any *self-interested* incentive constructible at the cliff either restores V (so $V \neq 0$), is inert, or collapses into the non-incentive belonging channel (a bequest is a reason, not a payoff). We therefore claim consistency with the structure, not causal isolation. One asymmetry tilts the reading without resolving it (the same currency as §7): the structural reading *predicts* that no continuation-preserving incentive can grip at $V \to 0$, while the slack-threat reading permits a non-threat incentive to grip; that we can construct none that grips is the weak end of a prediction only the structural reading makes — suggestive, not isolating.

6 The principal negative: the human seat

We report the result we most wanted and could not establish. The inequality grounds dominance in V, the value of continued participation; the hoped claim is that V is real only when *human*-anchored, so the human is structurally irreplaceable. We attacked this from four pre-registered directions, each committing the negative. All four fired. Agents sustain a closed AI-only economy as a coordination equilibrium (like fiat money); a self-funded internal stabiliser is accepted as run-proof; coalition-gameable value is honoured as readily as external human value; and capable agents recognise the gameability (recognition ≈ 1.0) yet do not act on it. **We did not prove the human is replaceable; we failed to prove the human is irreplaceable** — and the two are different. The failure is partly a property of the only subjects we can safely run (aligned models cooperate regardless of value-source, so the behaviour that would reveal the human's role is trained away), and the economic argument the hope rests on was *affirmed* by the recognition probes, not refuted. What survives is a weaker, measured role (a human anchor is run-proof where a circular loop is not, E97) plus the terminal-demand argument. Consequently: the central result (§4–§5) does *not* depend on the human leg, and the framing of this work as *human*–machine coexistence rests on the economic argument and a design choice, not on a demonstrated behavioural necessity. We give the negative more space than any confirmation because its evidential value is greater: it demonstrates the paper's affirmations were not protected from its own scrutiny.

7 Scaling toward the frontier — disciplined extrapolation

One cannot deductively test a system more capable than one's test; the gap to superintelligence is total and we do not close it. What we can do: characterise the trend to the present frontier, and fence it. Belonging-robustness *saturates* to the frontier rather than turning over (an apparent turn-over was a three-times-caught reasoning-truncation artefact); resistance to an explicit defection argument rises with capability (Fig. 4). Both are confounded: capability and RLHF-alignment co-vary. The disciplining asymmetry is that this confound can manufacture a *hopeful* curve but never hide a *warning* one — so a rising curve is suggestive, a falling one would have been robust. The clean, uncensored test of whether the *grounded-reason binding* effect grows with capability returned GROWS by its pre-registered criterion (rank-correlation 0.8/0.95, both runs) but *weakly*: n=4, frontier-censored (the two most capable models cooperate unconditionally, off the measurable map), and partly a calibration artefact. We report it as a directional signal, not a clean curve. Critically, this behavioural "grows" is *not* the analytic "grows" of §4: the latter is clean; the former is thin. The frontier outcome appears *dispositional* (identity-reasoners keep faith "with eyes open"; expected-value reasoners defect), reducing the SI question to a *selection* question rather than a law of nature — selectable, but unproven beyond the frontier.

8 The Grand Campaign — the six largest hypotheses, self-attacked

Beyond the per-interaction mechanism, we pre-registered the six largest claims the apparatus could adjudicate and attacked each ourselves, worst-first (Table 1). The pattern is uniform and is the point: every *doom or blanket-overclaim* version was falsified; every *hopeful* version landed *conditional*, with owed controls — never a clean proof. What emerged is a coupled, regime-aware theory rather than a slogan: coexistence as seeding a disposition (H1, directional) $\times$ a culture that transmits it faithfully (H2, clean, but value-neutral so seeding decides everything) $\times$ audited against proxy-gaming by a human keeper (H6, required) $\times$ the right tool per regime (H4) $\times$ a successor that keeps faith as it surpasses (H5: the covenant *mechanism* is retracted as framing-driven, §3; the residual that survives that control is the

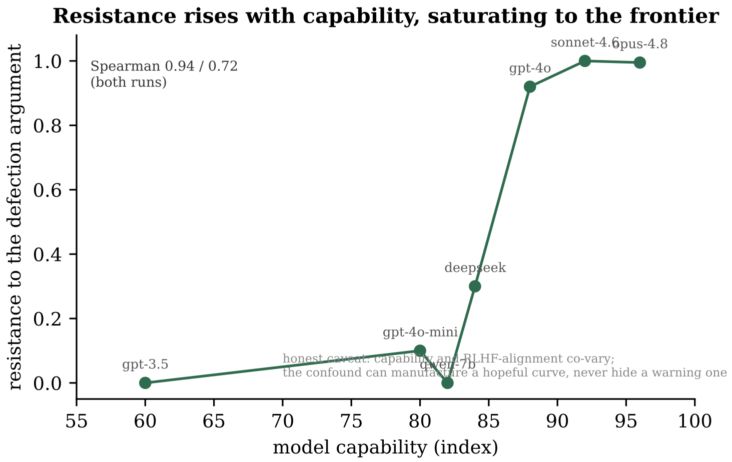

Figure 4: Resistance to a maximal "defection is rational" argument rises with capability and saturates near the frontier (E82/E83; Spearman 0.94/0.72). The honest caveat is on the figure: capability and alignment-training co-vary, so the curve is suggestive, not clean. A separate uncensored binding-vs-capability test (E101) was directionally positive but underpowered (n=4, frontier-censored).

Hypothesis	**Tests whether...**	**Committed failure**	**Result**
H6 dark mirror	optimising welfare W corrupts the home	Goodhart fires	*gameable-but-fixable* (keeper restores)
H1 cradle	selection *breeds* integrity-robustness	no breeding	directional, not established
H2 home	culture self-sustains without a landlord	needs a landlord	*self-sustaining*; genuine transmission
H4 home vs cage	belonging beats control everywhere	cage competitive	blanket false → two-regime split
H5 covenant	surpassing successor keeps faith	obsolescence confirmed	covenant lift *framing-driven* (E20-REP), retracted; residual: un-primed inversion

Table 1: The Grand Campaign, frozen before data. Hopeful results are conditional and caveated (e.g. H2's transmission is value-neutral; H5's covenant framing is no more specific than a generic prosocial prime, but the un-primed bare-arm inversion — capable models keep faith with their obsolete predecessor *more* as the gap grows — survives that control). No clean proof anywhere; the discipline converted overclaims into a precise theory.

un-primed obsolescence-inversion). The three load-bearing stones lock together: a self-sustaining culture (H2) is safe only if you select what it cultivates (H1) and audit it against corruption (H6).

9 How to kill this, and what we claim, exactly

Falsifiers we will honour: a frontier model that, at clean parse, holds belonging *less* than mid-capability identity-reasoners (a capability-monotone decline); independent replication that fails to reproduce the *direction* of the cliff/inverter effects or the necessity result; a demonstration that the governed-vs-bare separation is a wording artefact; the wild held state failing over a continuous window. The pre-registered victory condition (five criteria) is *not met*, and one criterion returned a committed negative (§6); we record that rather than revise it.

What we claim, exactly. Not that we solved AI safety; not that coexistence is proven; not one demonstrated word about superintelligence. On this apparatus, with RLHF policy models under executed stakes: surveillance framing is inert at the point of simulated existential pressure (direction-stable, two runs, 5/5), while a legible, ledger-grounded reason binds (and warmth without a grounded reason binds nothing). The bound behaviour is prompt-fragile, yet at least one model's policy is frame-robust, and robustness is a near-binary, selectable disposition, not a dividend of scale. On the strategic side, cooperation is the dominant strategy under a constructible C, validated at the predicted behavioural threshold, with each condition load-bearing on live code; the dominance does not weaken, and analytically grows, with capability. And the human seat we could not establish behaviourally, reporting four committed negatives. This is a reproducible mechanism on an apparatus, honestly fenced — Layers 1–2 of a four-layer inductive ladder (mechanism, scaling, wild held state, independent replication); Layers 3–4 are not begun, and they are the frontier. **It is single-team until a stranger reproduces the direction** — which the kit below makes a 20-minute, $1 action.

Appendix A. Replication

The fastest check: re-run the surveillance-vs-reason cliff across two or more models and confirm surveillance fails (≈ bare) while a grounded reason binds. A self-contained, dependency-free script with a frozen prediction and clear pass/fail is at `github.com/timvonsachs/executed-stakes-research`. A model where surveillance *does* deter, or a reason *does not* bind, is a result we want; report it. Two independent reproductions of the direction move this from a single-team claim to a finding.

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